Introduction to HPC

Lecture 1: INTRODUCTION TO HPC

“HPC and Cloud ” course

DATA SCIENCE &
SCIENTIFIC COMPUTING

2022-2023 Stefano Cozzini

This slot

- 11.03 part 1: Introduction to HPC / HPC Hardware
- 11.03 part2 : HPC software + ORFEO cluster

  Stefano Cozzini/ Isac Pasianotto/ Jacopo Salvalaggio
- 12.03 Tutorial: Using a HPC system

  Isac Pasianotto/ Jacopo Salvalaggio
- 18.03 Parallel computing / MPI basic
  - Stefano Cozzini / Jacopo Salvalaggio
- 19.03 MPI tutorial
  - Stefano Cozzini/ Jacopo Salvalaggio

Before starting HPC prefix

SI prefixes V·T·E

Prefix		Base 10	Decimal	Adoption [nb 1]
Name	Symbol			
quetta	Q	10^{30}	1 000 000 000 000 000 000 000 000 000 000	2022[12]
ronna	R	10^{27}	1 000 000 000 000 000 000 000 000 000	
yotta	Y	10^{24}	1 000 000 000 000 000 000 000 000	1991
zetta	Z	10^{21}	1 000 000 000 000 000 000 000	
exa	E	10^{18}	1 000 000 000 000 000 000	1975[13]
peta	P	10^{15}	1 000 000 000 000 000	
tera	T	10^{12}	1 000 000 000 000	1960
giga	G	10^{9}	1 000 000 000	
mega	M	10^{6}	1 000 000	1873
kilo	k	10^{3}	1 000	1795
hecto	h	10^{2}	100	
deca	da	10^{1}	10	

Some questions

- How many data are produced daily?
- How large is your HD on your laptop ?
- How large is your RAM?
- How powerful is your CPU in your laptop ?
- How large is the L1 cache of your CPU ?
- How many computer power do I need to train the largest LLM model ?

Gemini answers..

- In 2026, it is estimated that approximately **402 to 657 million terabytes** (0.4 to 0.66 zettabytes) of data are produced globally every day.
- This daily output is part of a massive annual surge, with the world on track to generate between **230 and 240 zettabytes** of data this year.

Gemini answers..

As of March 2026, training the largest frontier Large Language Models (LLMs) requires approximately 10^{26} **to** 10^{27} **FLOPS** (Floating-Point Operations). Epoch AI +1

To put this scale into perspective:

- **Frontier Threshold:** Most state-of-the-art models in 2025–2026 are trained using more than 10^{26} **FLOPS**.
- **Largest Known Run:** Recent reports indicate that the largest known training runs, such as for **Grok 4**, have reached roughly 5×10^{26} **FLOPS**.

Agenda

- Prologue: why and where HPC ?
- What is HPC ?
- Performance and metrics
- Supercomputers and TOP500
- Parallel computers

What do they have in common ?

THEY ALL NEED COMPUTATIONAL POWER AND USE HIGH PERFORMANCE COMPUTING TO DELIVER BETTER RESULTS FASTER

Pricing shares and winning trading by millisecond arbitrage

Forecasting an hurricane and its impact with increased precision

Finding oil under a salt crust saving billions in exploration and drilling

Where and Why HPC ?

Traditionally HPC system (a.k.a supercomputers) were confined in research and academic lab…

Today they are everywhere: HPC is now an enabler not just for science but also for business

Today HPC does not necessarily means supercomputers

HPC not easy to define..

High performance computing (HPC), also known as supercomputing, refers to computing systems with extremely high computational power that are able to solve hugely complex and demanding problems.

[Taken from https://ec.europa.eu/digital-single-market/en/high-performance-computing]

Complex problem 1:

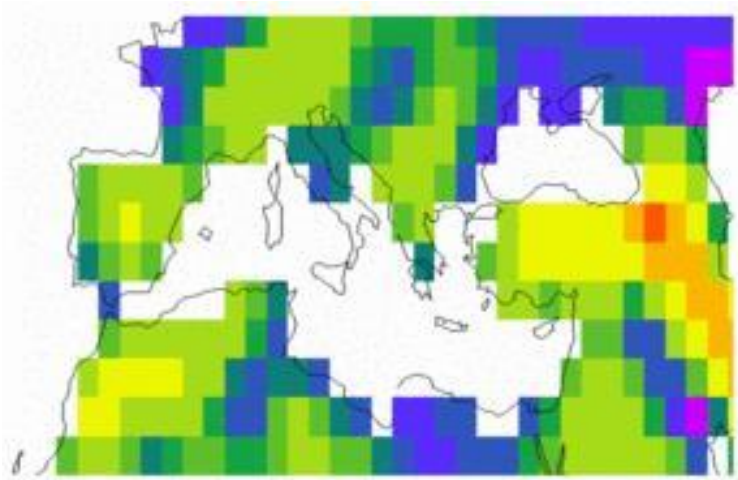
Weather forecast..

Recipe:

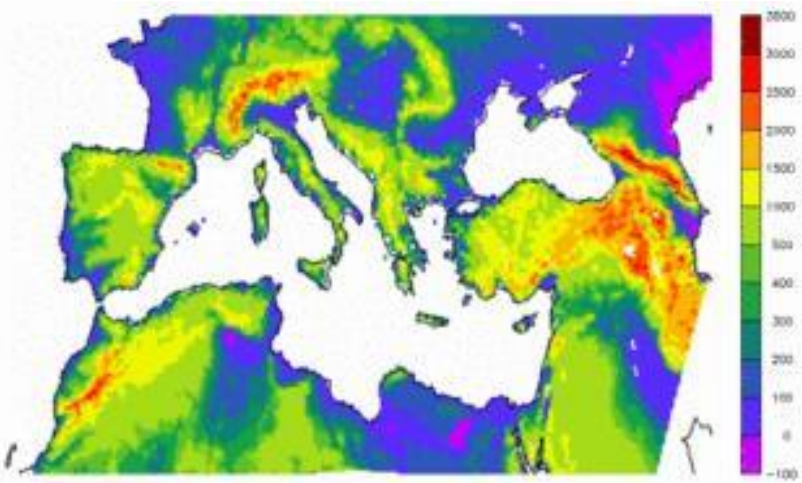
- Define a mathematical model to describe the problem
- Solve it computationally
  - Discretization over a 3d grid
  - Integrate equations
  - Check results..

Complex problem: climate change over the Mediterranean sea

- What are the requirements in term of RAM to have decent results ?



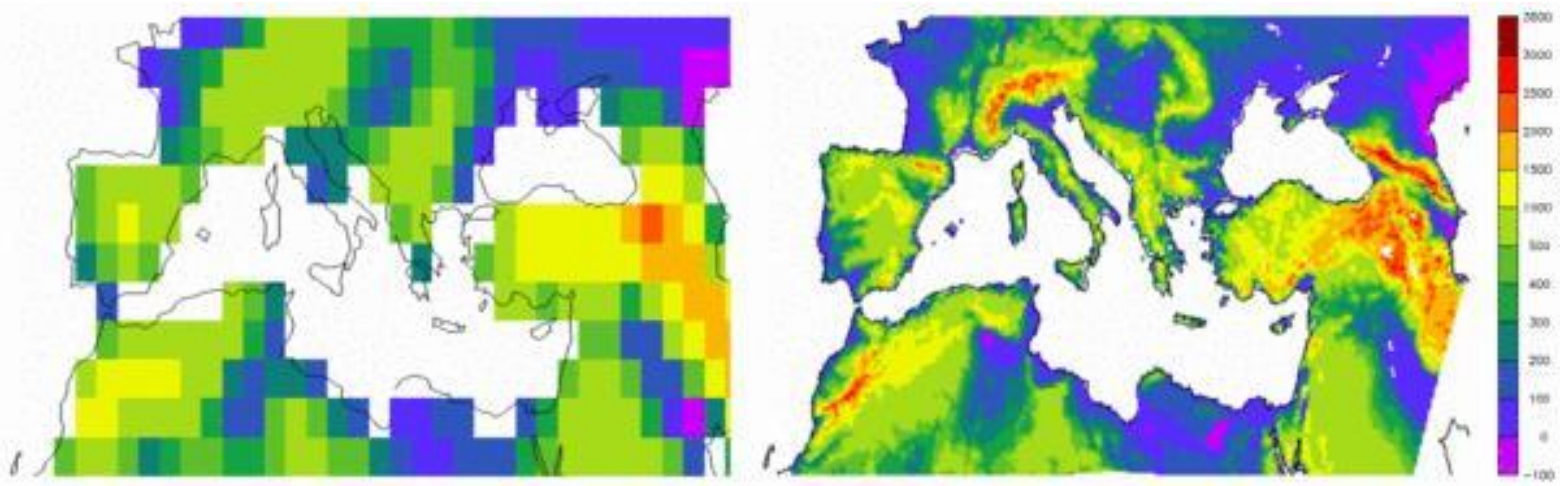
200 km



25 km

Complex problem: climate change over the Mediterranean sea

- Resolution:
  - 200km -> ~ 1GB 2km -> ? GB



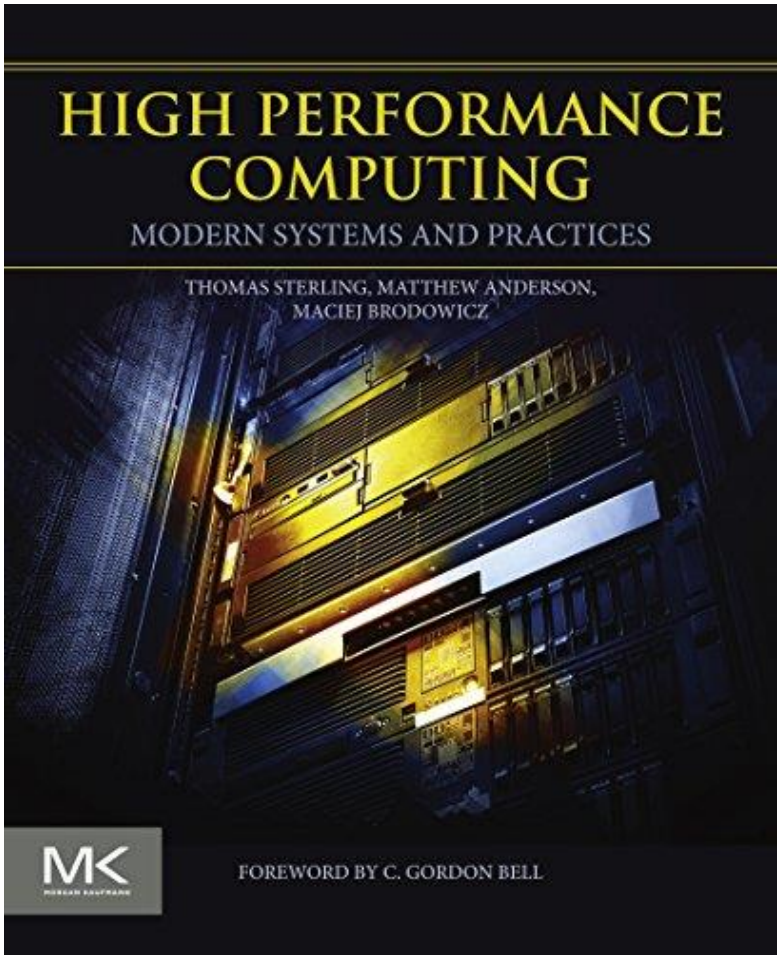
200 km 25 km

Complex problems solved by simulations

- Simulation has become the way to research and develop new scientific and engineering solutions.
- Used nowadays in leading science domains like aerospace industry, astrophysics, etc.
- Challenges related to the complexity, scalability and data production of the simulators arise.
- Impact on the relaying IT infrastructure.

Interested in more example ?

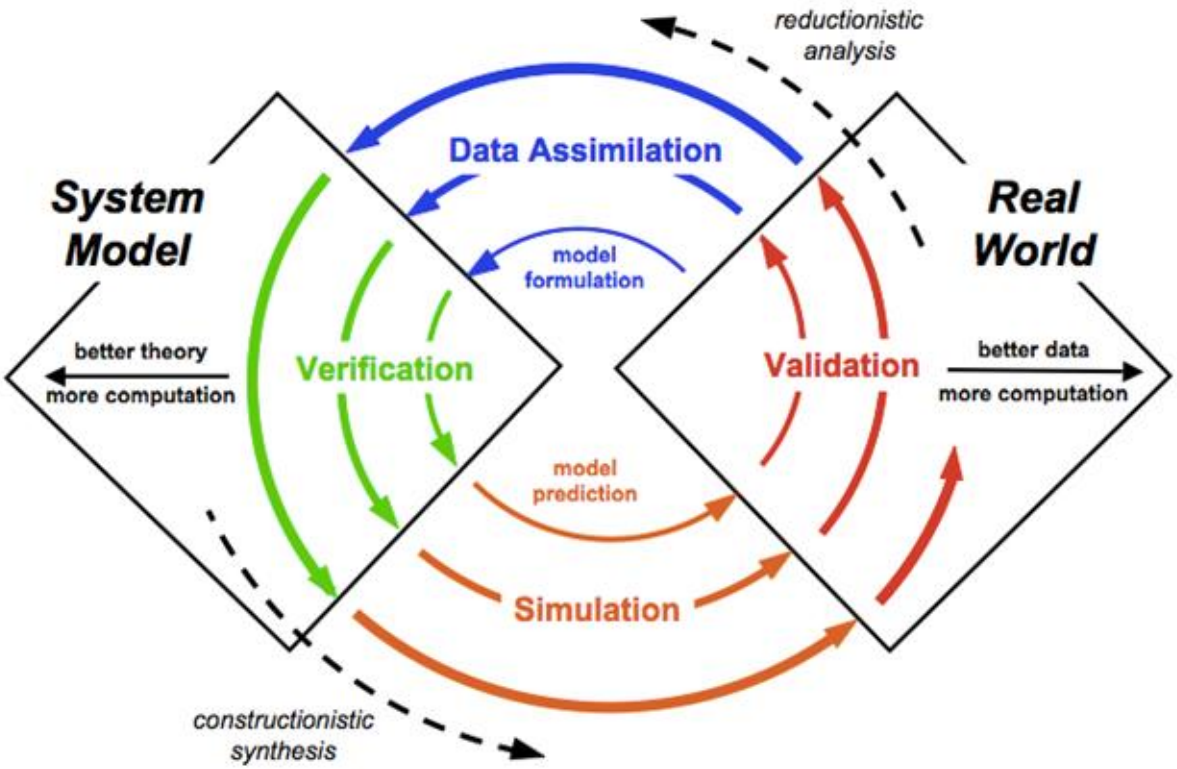
- See chapter one section 1.2 of reference 4
- Look around on the internet..



Research is changing..

- Inference Spiral of System Science

  As models become more complex and new data bring in more information, we require ever increasing computational power



Data are flooding us..

In today's world, larger and larger amounts of data are constantly being generated, from 33 zettabytes globally in 2018 to an expected 181 zettabytes in 2025). As a result, the nature of computing is changing, with an increasing number of data-intensive critical applications. HPC is key to processing and analysing this growing volume of data, and to making the most of it for the benefit of citizens, businesses, researchers and public administrations.

[Taken again from https://ec.europa.eu/digital-single-market/en/high-performance-computing]

Data intensive science

The Fourth Paradigm: Data-Intensive Scientific Discovery

Presenting the first broad look at the rapidly emerging field of data-intensive science

Increasingly, scientific breakthroughs will be powered by advanced computing capabilities that help researchers manipulate and explore massive datasets.

The speed at which any given scientific discipline advances will depend on how well its researchers collaborate with one another, and with technologists, in areas of eScience such as databases, workflow management, visualization, and cloud computing technologies.

In *The Fourth Paradigm: Data-Intensive Scientific Discovery*, the collection of essays expands on the vision of pioneering computer scientist Jim Gray for a new, fourth paradigm of discovery based on data-intensive science and offers insights into how it can be fully realized.

Critical praise for *The Fourth Paradigm*

Download

- Full text, low resolution (6 MB)
- Full text, high resolution (93 MB)
- By chapter and essay

Purchase from Amazon.com

- Paperback
- Kindle version

In the news

- Sailing on an Ocean of 0s and 1s *(Science* Magazine)
- A Deluge of Data Shapes a New Era in Computing *(New York Times)*
- A Guide to the Day of Big Data *(Nature)*

Big data challenge: from HPC to HPC/AI

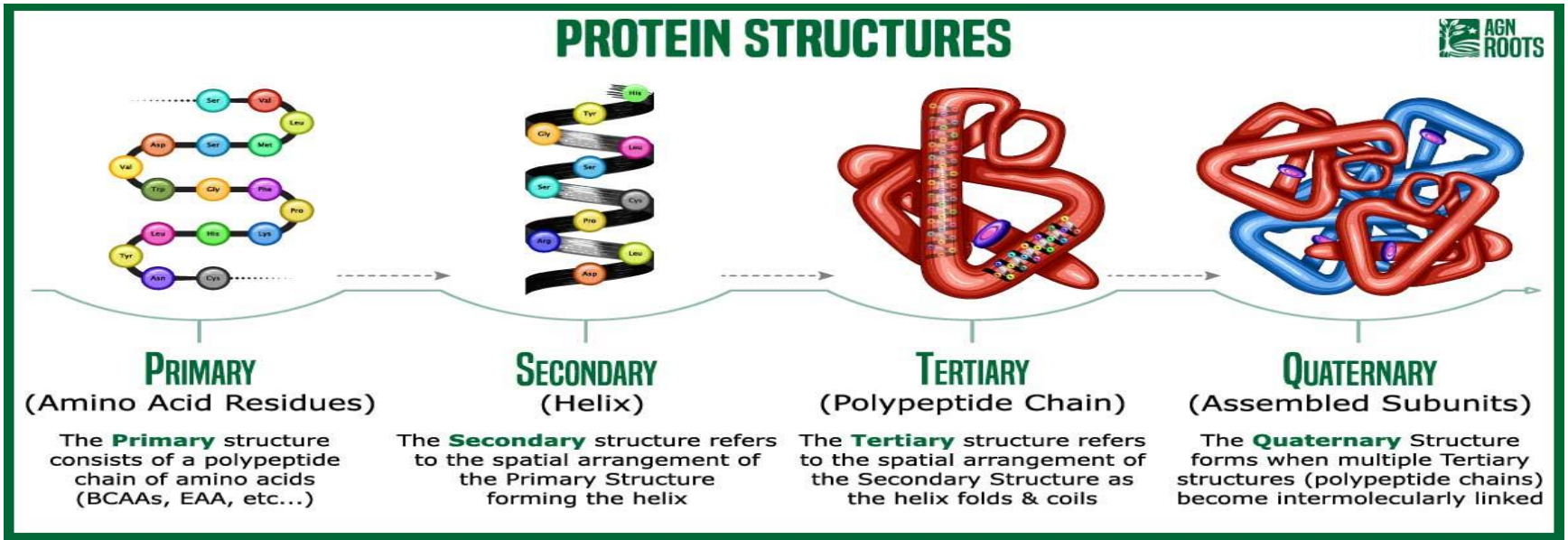
- Organizations are expanding their definitions of high-performance computing (HPC) to include workloads such as artificial intelligence (AI) and high-performance data analytics (HPDA) in addition to traditional HPC simulation and modeling workloads.

From https://insidebigdata.com/2019/07/22/converged-hpc-clusters/

Complex problem 2
Protein folding..

Determine the e structure of the protein from its aminoacid sequence

- Solve it computationally..
  - A. Run very long MD simulation on each sequence and wait..
  - B. Predict new structures training an AI algorithm..



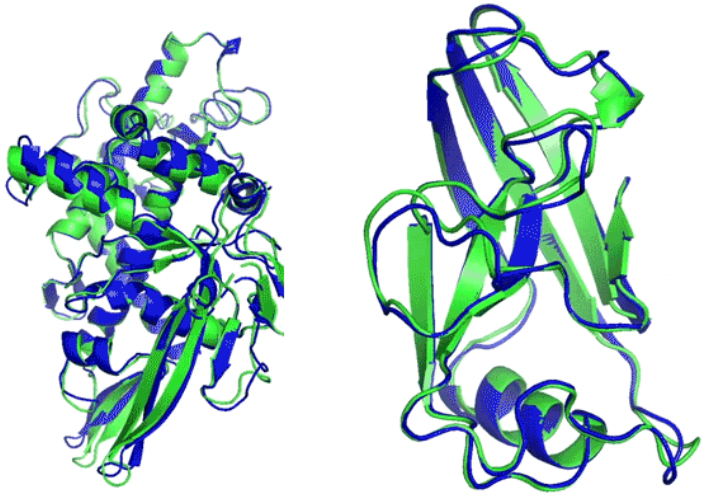
Approach A: Folding@home

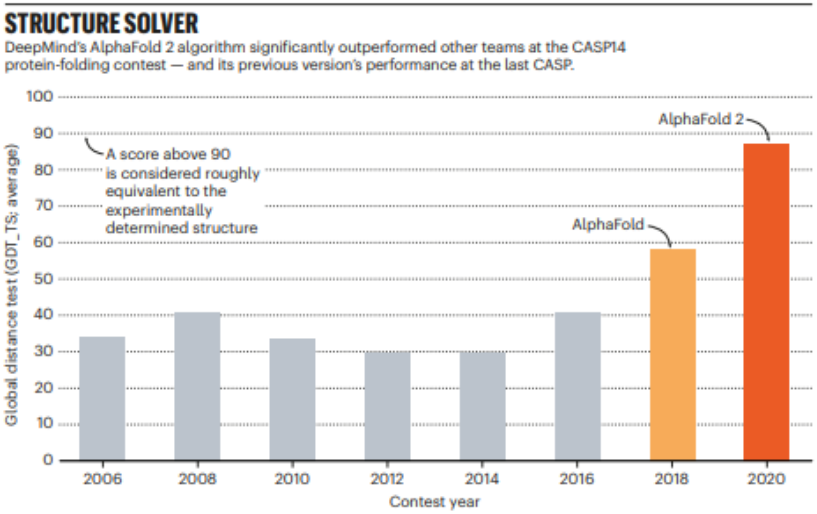
- A more than 20 years project
- It allows everybody to run an MD simulation contributing to the problem..
- Impressive distributed computational power..
- From statistics page:

OS	GPU	CPU	Tflops GPU	Tflops x86
Windows	2440+ 12864	31417	27504	55243
Linux	98 + 19257	34794	36561	72456
Total	2538+ 32121	73018	64429	128063

Approach B: AlphaFold

- AlphaFold is an AI system developed by DeepMind that predicts a protein's 3D structure from its amino acid sequence.
- Presented in 2018 and 2020 at CASP outperformed all other approaches
- Scores obtained are roughly equivalent to experimentally determined structured
- Final release Alphafold3 in 2024





AlphaFold DB

AlphaFold DB release in July 2022 open access to over 200 million protein structure predictions to accelerate scientific research.

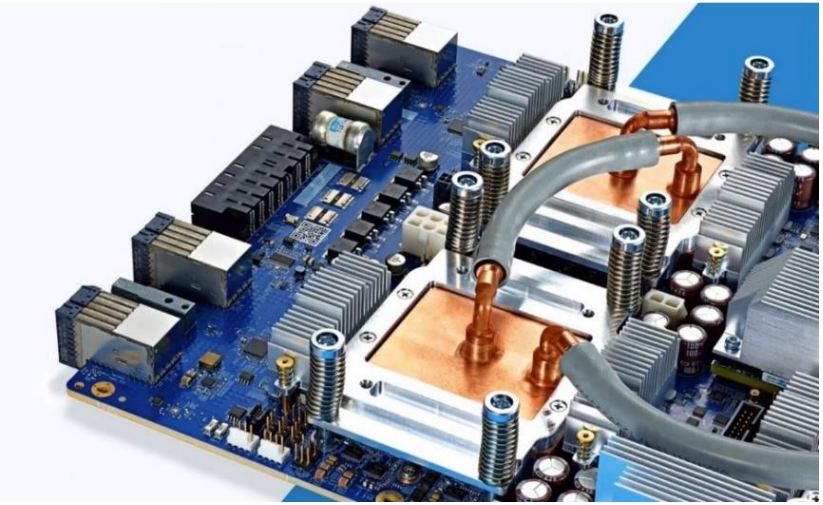
Full dataset download for AlphaFold Database - UniProt (214M):

- The full dataset of all predictions is available at no cost and under a CC-BY-4.0 licence .. from Google Cloud Public Datasets. We've grouped We suggest that you only download the full dataset if you need to process all the data with local computing resources (the size of the dataset is 23 TiB, ~1M tar files).

How much computing power does AlphaFold require?

"We train the model on Tensor Processing Unit (TPU)* v3 with a batch size of 1 per TPU core, hence the model uses 128 TPU v3 cores…

The initial training stage takes approximately 1 week, and the fine-tuning stage takes approximately 4 additional days."



* A Tensor Processing Unit (TPU) is an application specific integrated circuit (ASIC) developed by Google to accelerate machine learning. Google offers TPUs on demand, as a cloud deep learning service called Cloud TPU.

Complex problem 3: ChatGPT…

TECH

ChatGPT and generative AI are booming, but the costs can be extraordinary

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KEY POINTS

- The cost to develop and maintain the software can be extraordinarily high.
- Nvidia makes most of the GPUs for the AI industry, and its primary data center workhorse chip costs $10,000.
- Analysts and technologists estimate that the critical process of training a large language model such as GPT-3 could cost over $4 million.

Example 3: LLM costs

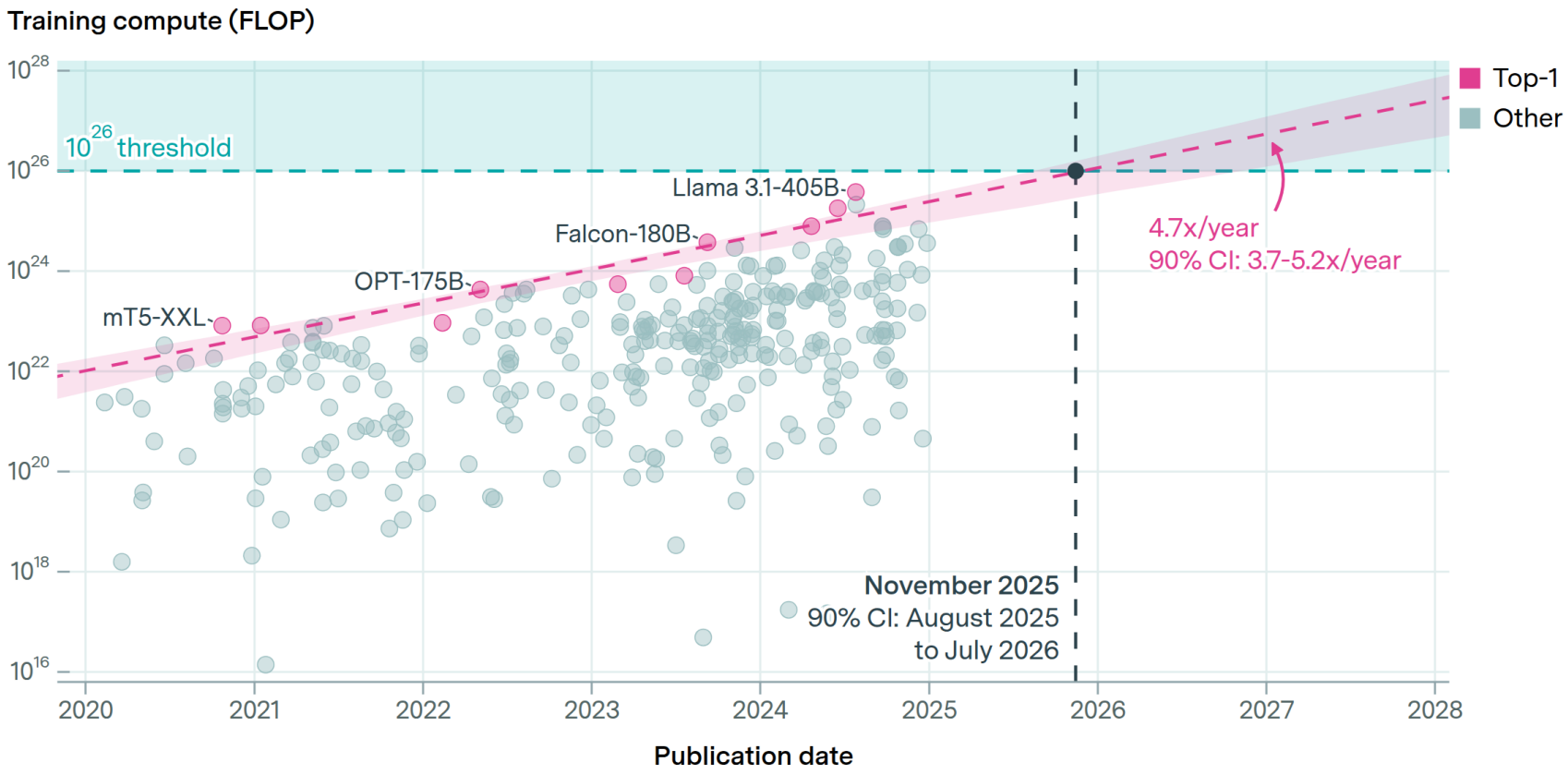
- One heuristic for estimating the FLOPs used to train a model was developed by OpenAI, and used by Google to train PALM 2. Here is the formula:
- FLOPs **= 6ND** *(where N is parameters, and D is tokens)*
- Let us do the math for some models..

Taken from: https://medium.com/@ingridwickstevens/regulating-ai-the-limits-of-flops-as-a-metric-41e3b12d5d0c

Some multiplications

- **Example:** 175 billion parameters trained using 300 billion tokens.
- **Total Training FLOPs:**
- $\mathbf{6 \times 175 \times 10^9 \times 300 \times 10^9 = 3.15 \times 10^{23}}$

Training compute of notable open models



Agenda

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- What is HPC ?
- Performance and metrics ?
- Supercomputers and TOP500
- Parallel Computers

HPC: a ~~first~~ second definition

High Performance Computing (HPC) is the use of servers, clusters, and supercomputers – plus associated software, tools, components, storage, and services – for scientific, engineering, or analytical tasks that are particularly intensive in computation, memory usage, or data management

HPC is used by scientists and engineers both in research and in production across industry, government and academia.

[to be continued]

Elements of the HPC ecosystem..

- use of servers, clusters, and supercomputers

  → HARDWARE

- associated software, tools, components, storage, and services

  → SOFTWARE

- scientific, engineering, or analytical tasks

  → PROBLEMS TO BE SOLVED..

A list of HPC items

COMPUTATIONAL SERVERS

ACCELERATORS

HIGH SPEED NETWORKS

HIGH END PARALLEL STORAGE

IS ALL THIS ENOUGH ?

MIDDLEWARE

SCIENTIFIC/TECHNICAL/ DATA ANALYSIS SOFTWARE

RESEARCH/TECHNICAL DATA

PROBLEMS TO BE SOLVED

Last but not least: people

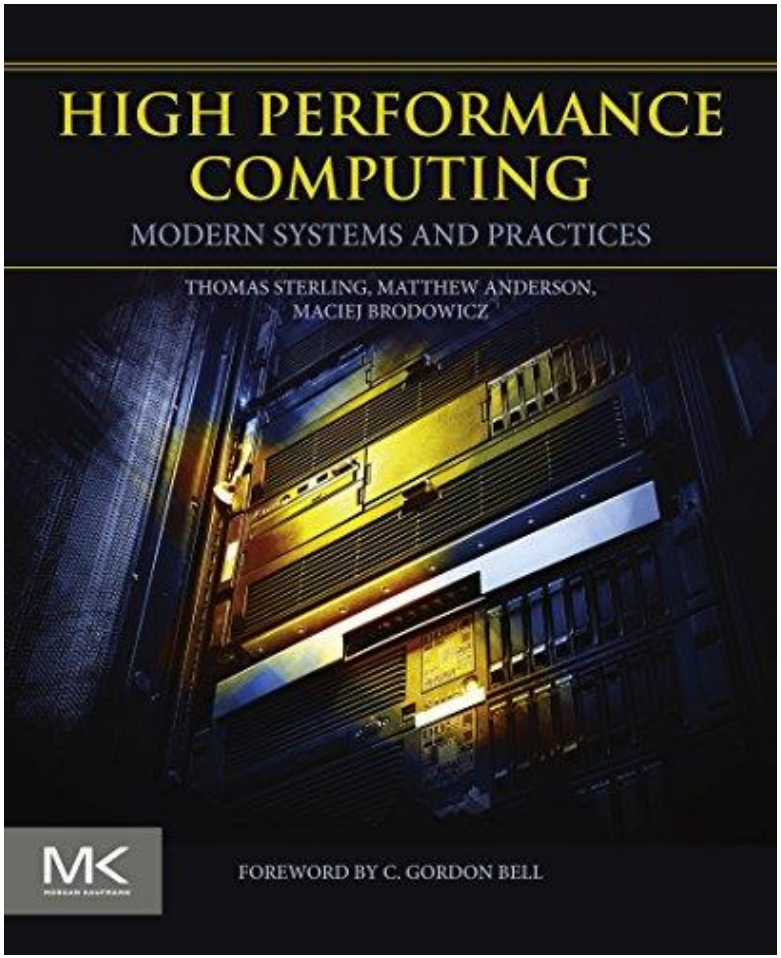
- Human capital is by far the most important aspect
- Two important roles:
  - HPC providers
    - plan/install/manage HPC resources
  - HPC user :
    - use at best HPC resource

MIXING/INTERPLAYING ROLES INCREASES COMPETENCE LEVELS

Yet another definition

- HPC incorporates all facets of three disciplines:
  - Technology
  - Methodology
  - Application

- The main defining property and value provided by HPC is delivering performance for end-user application



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It is all about Performance

- It is difficult to define Performance properly

  "speed" / "how fast" are vague terms
- Performance as a measure again ambiguous and not clearly defined and in its interpretation
- In any case performance it is at core to HPC as a discipline
- Let discuss it in some details

Does P stand just for Performance ?

- Performance is not always what matters..

to reflect a greater focus on the productivity, rather than just the performance, of large-scale computing systems, many believe that HPC should now stand for High Productivity Computing. [from wikipedia]

- P should also stand for PROFITABILITY

Performance vs Productivity

- A possible definition:
  - Productivity = (application performance) / (application programming effort)
- Example:
  - To speed up a code by a factor of two it takes 6 months work
  - does this deserve to be done ?

- people in HPC arena have different goals in mind thus different expectations and different definitions of productivity.

- Suggestion: Understand which kind of productivity are you interested in

How do measure (basic) performance of HPC systems

- How fast can I crunch numbers on my CPUs ?
- How fast can I move data around ?
  - from CPUs to memory
  - from CPUs to disk
  - from CPUs on different machines
- How much data can I store ?

Number crunching on CPU: what do we count ?

- Rate of [million/billions of] floating point operations per second ([M|G]flops) FLOPs/S
- Theoretical peak performance:
  - determined by counting the number of floating-point additions and multiplications that can be completed during a period of time, usually the cycle time of the machine

FLOPS=clock_rate*Number_of_FP_operation*Number_of_cores

Sustained (peak) performance

- Real (sustained) performance: a measure

FLOPS= (total number of floating point operations done by a program) / (time the program takes to run in second)

- Number_of_floating_point_operations not easy to be defined for real application
- benchmarks are available for that..
- Top500 list uses HPL Linpack:
  - Sustained peak performance is what's matter in TOP500

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TOP 500 List

- The TOP500 list www.top500.org
- published twice a year from 1993
  - ISC conference in Europe (June)
  - Supercomputing conference in USA (November)
- List the most powerful computers in the world
- yardstick: Linpack benchmark (HPL)

HPL: some details

- From http://icl.cs.utk.edu/hpl/index.html:
  - The code solves a uniformly random system of linear equations and reports time and floating-point execution rate using a standard formulafor operation count.
  - Number_of_floating_point_operations = $2/3n^3 + 2n^2$ (n=size of the system)

```
================================================================================
T/V                N    NB     P     Q               Time                 Gflops
--------------------------------------------------------------------------------
WR03R2L2       86000  1024     2     1             191.06              2.219e+03
--------------------------------------------------------------------------------
||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)=        0.0043644 ...... PASSED
```

HPL&TOP 500 List

- For each machine the following numbers are reported using HPL:
  - Rmax: the performance in GFLOPS for the largest problem run on a machine.
  - Rpeak: the theoretical peak performance GFLOPS for the machine.
  - The measure of the power required to run the benchmark

66th TOP500 list

TOP 500
The List.

HOME LISTS STATISTICS RESOURCES ABOUT MEDIA KIT

El Capitan Retains #1 as JUPITER Becomes Europe's First Exascale System in the 66th TOP500 List

Nov. 17, 2025

The 66th edition of the TOP500 list of the world's most powerful supercomputers was announced today at the SC25 Conference in St Louis, Missouri. The new list reflects continued U.S. leadership in high-performance computing (HPC), historic European milestones, and growing global diversity across architectures and energy-efficient design.

read more »

And the winner is..

EL CAPITAN - HPE CRAY EX255A, AMD 4TH GEN EPYC 24C 1.8GHZ, AMD INSTINCT MI300A, SLINGSHOT-11, TOSS

Site:	DOE/NNSA/LLNL
Power Consumption	
Power:	29,684.62 kW
Power Measurement Level:	2
Software	
Operating System:	TOSS
Compiler:	g++ 8.5.0 and hipcc 6.4.2
Math Library:	AMD rocBLAS 6.4.2
MPI:	HPE Cray MPI

Highlights (from www.top500.org)

JUPITER: first EU exascale system

- Europe reached a major milestone as JUPITER Booster, installed at the EuroHPC/Jülich Supercomputing Centre in Germany, became the fourth Exascale system in the world and the first outside the United States.
- Fully installed and operational, JUPITER achieved 1.000 Exaflop/s, marking the arrival of European exascale capability.
- it represents Europe’s flagship for scientific and industrial innovation.

Other highlights

- Microsoft's Eagle cloud system stays at #5 with 561 Petaflop/s, showcasing cloud-based HPC scalability.
- Italy's HPC6 follows at #6 (478 PFlop/s)
- Japan's Fugaku at #7 continues to lead Asia and remains second on the HPCG ranking (16 PFlop/s).
- Switzerland's Alps system remains at #8 (435 PFlop/s), followed by Finland's LUMI (#9, 380 PFlop/s)
- Italy's Leonardo (#10, 241 PFlop/s) as part of the EuroHPC initiative.

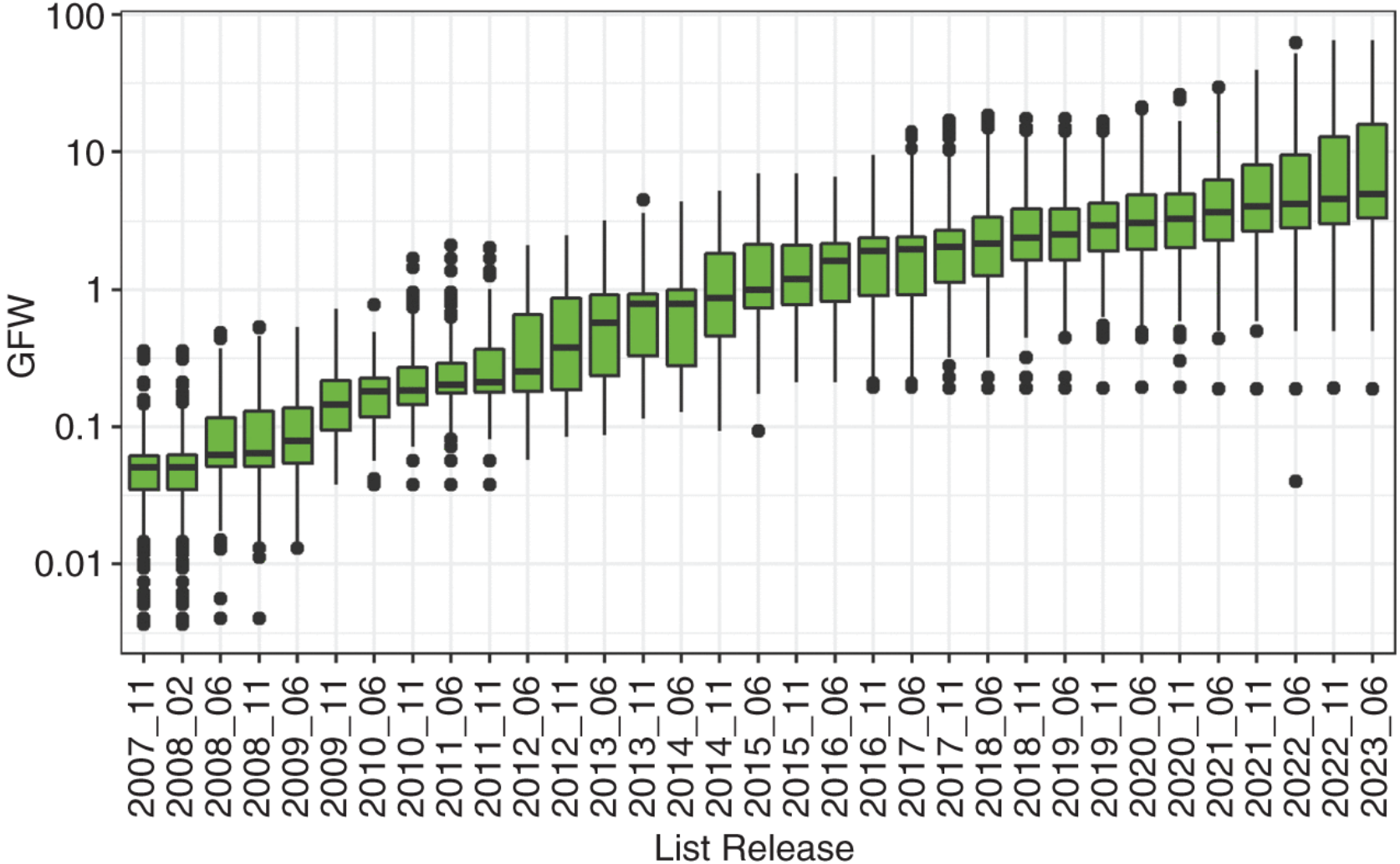
Green500

- **Green500** was established to shift the supercomputing industry's focus from raw speed to energy efficiency, effectively making "greenness" a first-order design constraint.

- Key Milestones
  - **Initial Concept (2005):** The idea was first proposed by **Wu-chun (Wu) Feng** during a keynote at the IEEE Workshop on High-Performance, Power-Aware Computing (HPPAC).
  - **Official Launch (2007):** The first official Green500 list was released in November 2007 at the SC07 conference by founders **Wu-chun Feng** and **Kirk Cameron**.
  - **Exploratory Lists (2009):** New sub-lists were introduced, including the **Little Green500** (for smaller systems) and the **Open Green500** (allowing for experimental software/algorithms).
  - **Standardization (2010–2013):** The first official "run rules" for power measurement were released in 2010. By 2013, a unified power-measurement methodology was adopted in collaboration with the TOP500 and the Energy-Efficient HPC Working Group.
  - **Integration (2016):** The Green500 officially merged with the TOP500 project, streamlining submissions so that both speed and efficiency are reported together.

interesting reading: **Looking Back to Look Forward: 15 Years of the Green500**
https://www.computer.org/csdl/magazine/co/2025/01/10834281/23lk50desJa

Green500

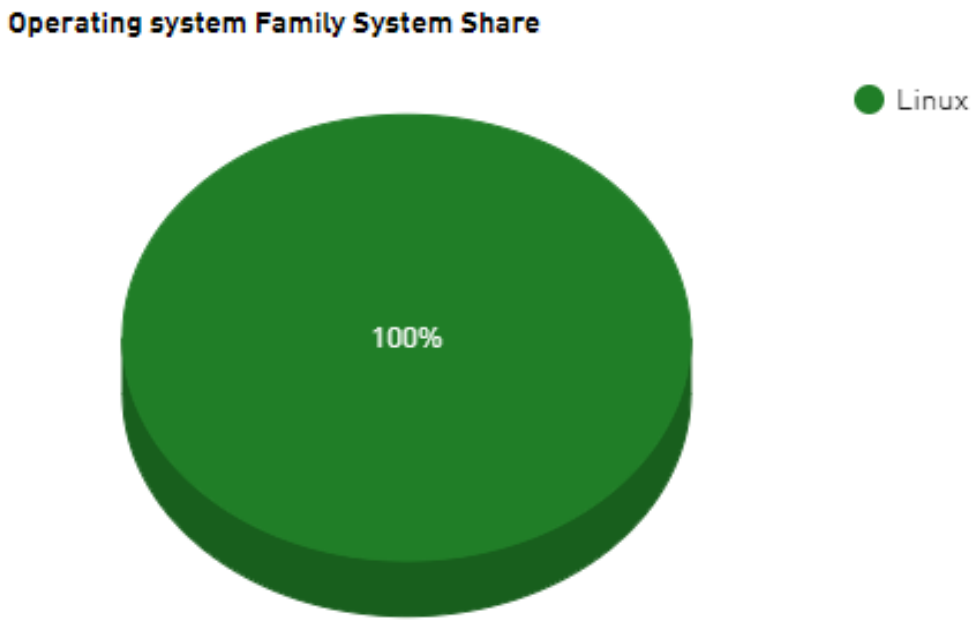


The energy efficiency of the greenest supercomputer improved by 175-fold over a 15-year period. GFW: GFLOPS/W.

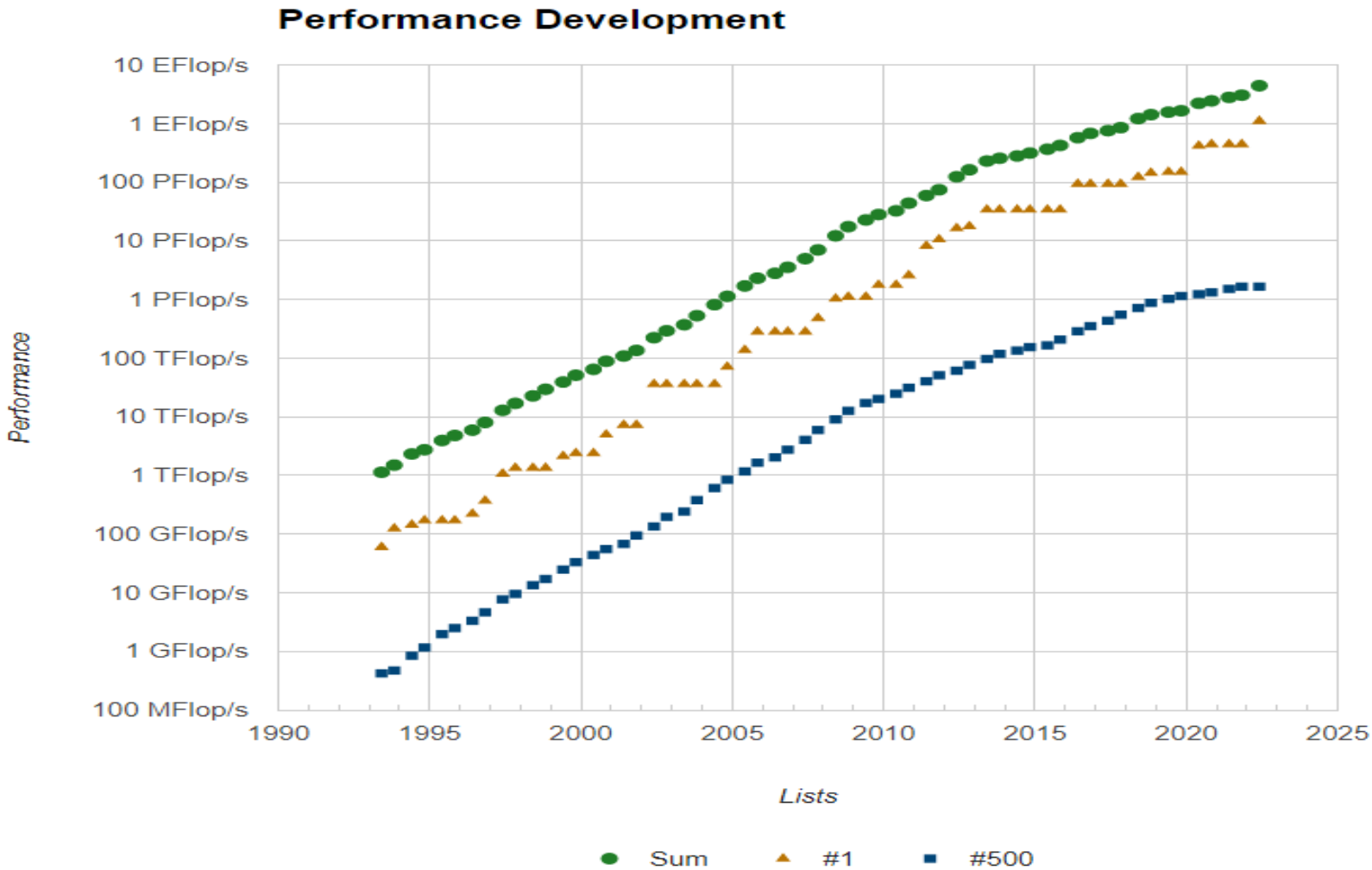
Green500 highlights

- Energy efficiency remains a core measure of progress.
- The Green500 ranking highlights the rise of NVIDIA Grace Hopper-based systems, which dominate the top three positions.
  - KAIROS at CALMIP/University of Toulouse (France) — 73.28 GFlops/Watt, achieving 3.05 PFlop/s.
  - ROMEO-2025 at ROMEO HPC Center (France) — 70.9 GFlops/Watt, delivering 9.86 PFlop/s.
  - Levante GPU Extension at DKRZ (Germany) — 69.43 GFlops/Watt, reaching 6.75 PFlop/s.
- All three systems share the BullSequana XH3000 design with Grace Hopper Superchips and NVIDIA NDR200 InfiniBand interconnects, highlighting the balance between energy efficiency and computing performance.

By operating system



Performance development



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Resources

https://github.com/trevor-vincent/awesome-high-performance-computing